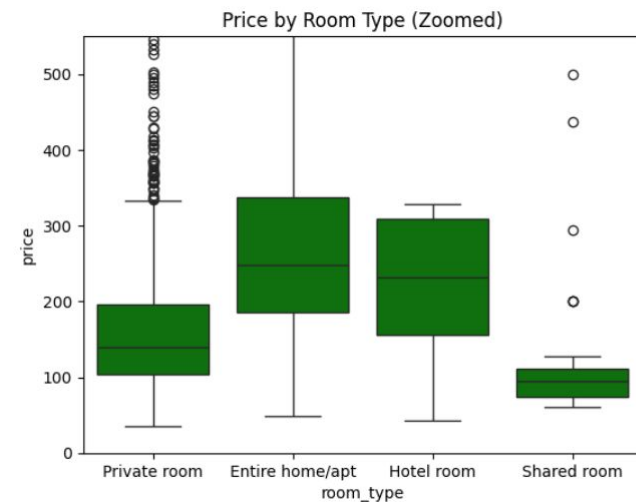
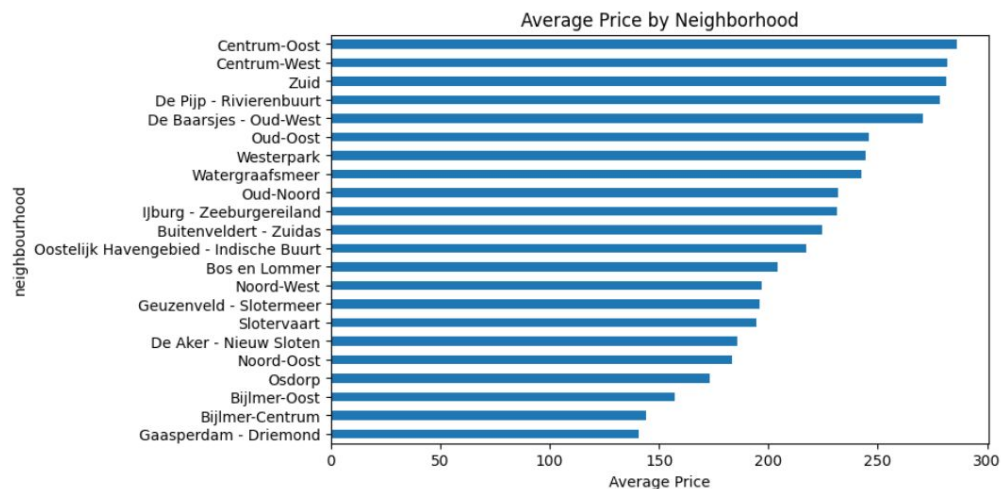


What factors determine Airbnb listing prices in Amsterdam?

Maya Beresteanu



Uncontrolled model:

$$\text{Price} = \beta_0 + \beta_1 * \text{Reviews}_i + \varepsilon_i$$

Limitations: Should interpret with caution due to skewed price data and heteroskedasticity, which may affect the precision of the estimates.

OLS Regression Results

Dep. Variable:	price	R-squared:	0.055
Model:	OLS	Adj. R-squared:	0.054
Method:	Least Squares	F-statistic:	85.40
Date:	Thu, 30 Apr 2026	Prob (F-statistic):	1.28e-70
Time:	13:56:45	Log-Likelihood:	-52774.
No. Observations:	5874	AIC:	1.056e+05
Df Residuals:	5869	BIC:	1.056e+05
Df Model:	4		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	351.3740	29.134	12.061	0.000	294.261	408.487
number_of_reviews	-0.4969	0.167	-2.975	0.003	-0.824	-0.169
room_type_Hotel room	5738.0431	316.449	18.133	0.000	5117.686	6358.400
room_type_Private room	-78.5673	67.135	-1.170	0.242	-210.177	53.042
room_type_Shared room	-169.3816	353.917	-0.479	0.632	-863.190	524.427

Omnibus:	15426.935	Durbin-Watson:	1.183
Prob(Omnibus):	0.000	Jarque-Bera (JB):	311503257.222
Skew:	30.722	Prob(JB):	0.00
Kurtosis:	1129.484	Cond. No.	2.52e+03

...

OLS Regression Results						
Dep. Variable:	price	R-squared:	0.001			
Model:	OLS	Adj. R-squared:	0.001			
Method:	Least Squares	F-statistic:	4.972			
Date:	Thu, 30 Apr 2026	Prob (F-statistic):	0.0258			
Time:	13:11:11	Log-Likelihood:	-52937.			
No. Observations:	5874	AIC:	1.059e+05			
Df Residuals:	5872	BIC:	1.059e+05			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	360.4574	27.991	12.878	0.000	305.585	415.330
number_of_reviews	-0.3480	0.156	-2.230	0.026	-0.654	-0.042
Omnibus:	15570.918	Durbin-Watson:	1.107			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	294936745.031			
Skew:	31.547	Prob(JB):	0.00			
Kurtosis:	1098.934	Cond. No.	194.			

Controlled model:

$$\text{Price} = \beta_0 + \beta_1 * \text{Reviews}_i + \beta_2 * \text{RoomType}_i + \varepsilon_i$$